



11 Physics 2017 Programme

WEEK	CONTENT	TEXT	PROBLEM SOLVING TEXT MANUAL		PRACTICALS	ASSESSMENT
Spring Term 3-5	Skill building/Working in Physics Units and measurement - prefixes, conversion. Scientific notation and significant figures. Scientific method / investigation process. Error and uncertainty. Graphing skills. Assignment 1 - Measurement.	Maths Skills 1, 2, 5-7 13-17 19, 20	Maths Skills 1, 2, 5-7 13-17 19, 20		Investigation - Density of paper	Prac Collection 1 Test: Measurement and Data
6-7	Vectors Vector and scalar quantities. Vector diagrams and representation. Addition and subtraction of vectors in one and two dimensions. Components of vectors.	6.1 6.2 6.3, 6.4 6.5	6.1 6.2 6.3, 6.4 6.5			
8-9	Motion Motion in one-dimension – definition of displacement, distance, velocity, speed, acceleration. Uniform acceleration – equations of motion. Motion under gravity. Graphical analysis of motion (one dimension) – slope, area under curve. Assignment 2 - Motion.	7.1 7.2, 7.4 7.5 7.3	7.1 7.2, 7.4 7.5 7.3	Set 14 Set 15	Expt 15.3 - Determining "g"	Prac Collection 2 Test: Motion
Summer Term 1-3	Forces Newton's First Law – inertia. Newton's Second Law – momentum, rate of change of momentum, impulse. Definition of force + free-body diagrams. Conservation of momentum – one dimension only, collisions, sports. Newton's Third Law - action and reaction. Distinction between mass and weight. Apparent weight in a moving reference frame.	8.3 8.1, 8.2 8.4, 8.6 8.5 8.7	8.3 8.1, 8.2 8.4, 8.6 8.5 8.7	Set 16 Set 17	Expt 16.1 - Newton's Second Law Expt 17.1 + 17.2 - Conservation of momentum	Evaluation and analysis - Airbags

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4-6	Work and Energy Distinction between work and energy. Calculation of work. Types of energy – potential (E_p) and kinetic (E_k) Energy transformations – efficiency. Conservation of energy. Power. Assignment 3 - Forces and energy.	9.1 9.2 9.3, 9.5 9.4, 2.1 9.6 9.7	9.1 9.2 9.3, 9.5 9.4, 2.1 9.6 9.7	Set 18		Test: Forces and Energy
7-10	Heating and Cooling Kinetic theory - solids, liquids, gases; potential energy (E_p) and kinetic energy (E_k). Heat versus temperature – difference. Temperature scales – Fahrenheit, Celsius, Kelvin. Specific heat – change of temperature Latent heat - change of phase, graphs of phase change with time. Method of mixtures calculations. Transferring heat – conduction, convection, radiation. Evaporation - factors affecting the rate, versus boiling, applications (body cooling, Coolgardie “safes”, reverse-cycle air conditioning, refrigerators). Degradation of energy into other forms. Assignment 4 - Heating and cooling.	1.1 1.2 1.3 1.4 2.2 - 2.4	1.1 1.2 1.3 1.4 2.2 - 2.4	Set 3 Set 4 Set 2	Expt 3.2 - Specific heat of a metal Expt 4.2 - Latent heat of ice	Prac. Test - specific heat of an unknown Test - Heating and cooling
Autumn Term 1-2	Electricity Electric charge. Current electricity - AC versus DC, conventional current versus electron flow.	5.1 5.2	5.1 5.2	Set 9		
3	Revision					
4-5	Semester 1 Examination					

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6-7	Electromotive force (EMF) and work done on moving charge. Potential difference and circuits - circuit symbols and diagrams, energy transformations. Resistance and resistivity. Ohm's Law – Ohmic versus non-Ohmic conductors.	5.3	5.3	Set 11	Invest 10.1 - Energy in a circuit	Prac. Test - value of unknown resistor
		5.4	5.4		Investigation - Ohm's Law	
8-9	Series and parallel circuits. Electric power. Household electricity. Safety with electricity - safety devices (earth wire, fuse, double insulation, earth leakage system). Assignment 5 - Electricity	5.5	5.5	Set 12	Investigation - Series and parallel circuits	Test - Electricity
		5.3	5.3	Set 10		
		5.5	5.5	Set 13		
10	Waves Types of waves - transverse and longitudinal. Graphing wave motion - displacement - distance, displacement - time. Wave equation - relationship between speed, wavelength and frequency.	10.1	10.1	Set 19		
		10.2	10.2			
Winter Term 1-2	Behaviour of waves - reflection and refraction. Wave interactions - superposition principle, resonance, standing waves in air columns and strings. Wave intensity - effect of distance from the source.	10.3	10.3	Set 20	Investigation - Reflection and refraction of light Invest 20.4 - Resonance in air columns	Prac Collection 3 Test - Waves
		10.4 10.5 10.6	10.4 10.5 10.6			
3-5	Nuclear Physics Atomic structure - atomic number, mass number, isotopes, radioisotopes. Types of radiation - alpha, beta and gamma; properties; symbols. Half-life and activity of radioisotopes - graphs, equation, applications (carbon dating). Uses of radioisotopes – medical, industrial, agricultural.	3.1	3.1	Set 5	Invest 5.2-5.4 - Radiation	
		3.2, 3.3 3.4	3.2, 3.3 3.4	Set 6	Invest 6.1 - Half-life	

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6	LIFE Week					
7-9	Biological effects of radiation - absorbed, dose, dose equivalent, quality factors.	3.5	3.5	Set 7		Evaluation and analysis - Fission and fusion Test - Nuclear Physics
	Nuclear binding energy - binding energy per nucleon, stability of nuclei, Einstein's equation, mass defect.	4.1	4.1	Set 8		
	Nuclear fission - chain reactions, critical mass, mass loss and energy release, weapons and reactors.	4.2	4.2			
	Nuclear fusion - solar reactions, fusion reactors, thermonuclear weapons. Assignment 6 – Nuclear Physics	4.3	4.3			
10	Revision					
	Semester 2 Examination					

References

Text: Physics Western Australia 11 (Pearson)
Manual: Exploring Physics – Year 11 (Science Teachers' Association of WA)
Practicals: Exploring Physics – Year 11 (Science Teachers' Association of WA)